

Sixinyang (Tom) Tian

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EDUCATION

Cornell University

Bachelor of Arts, Computer Science and Cognitive Science; Minor in Game Design

Ithaca, NY

Expected May 2028

GPA: 3.79

Relevant Coursework: Computer Graphics, Analysis of Algorithms, Machine Learning, Computer System Organization, Functional Programming, Computer Game Architecture, Operating Systems, Mathematical Foundations of Computing, Object-Oriented Programming & Data Structures

SKILLS

Languages: Java, Python, OCaml, C++/C/C#, Typescript/Javascript

Frameworks/Libraries: LibGDX, Playwright, Selenium, Pandas, PyTorch, NumPy

Tools: Git, Unity, Docker, Linux/UNIX, SPSS

Spoken Languages: English (Native), Mandarin Chinese (Native), Japanese (Elementary)

Additional: Video Editing, Singing/Viola/Music, Table Tennis

PROJECTS

BeeWitched (Steam) | Java, LibGDX, Figma

Jan 2026 – May 2026

Lead Programmer

- Architected and shipped a 2D Java game, leading a cross-functional team of 8 through authoring system design with UML activity diagrams and CRC cards, playtest integration, and release on Steam & finalist at Boston FIG 2026.
- Structured a modular class hierarchy across 8,700+ lines of production Java, establishing game objects and scenes to enable feature fine-tuning and speed up production.
- Implemented item sprites, inventory GUI, and text rendering with LibGDX's SpriteBatch, managing draw order and object lifecycles to prevent memory leaks; and built a persistent, rebindable input processing system.

OCamLiar's Bar | OCaml, Lwt, Unix

Mar 2025 – May 2025

Lead Programmer

- Built a real-time LAN multiplayer card-bluffing game in OCaml from scratch, authoring 74% of the 1,600+ line codebase and maintaining interface definitions, documentation, and automated testing with an OUnit2 test suite.
- Designed asynchronous client-server communication using Unix sockets and Lwt I/O, as well as an automatic public IP retriever through Google DNS, enabling low-latency gameplay across multiple terminals.
- Ensured correctness of program initiation and gameplay flow by utilizing finite state transitions and comprehensive error checking, keeping most modules around 60 lines for composability.

Interactive Image Selector | Java, Swing

Oct 2024 – Dec 2024

Programmer

- Implemented an "intelligent scissors" subject-masking tool that auto-traces image edges via Dijkstra's shortest-path algorithm over a pixel graph using a min-priority queue.
- Built live path previews and full undo/redo support in an MVC architecture, using mouse and keyboard listeners to drive real-time selection updates.
- Kept the GUI responsive with live progress bars and cancellation support by running Dijkstra's incrementally on a SwingWorker background thread.

EXPERIENCE

Laboratory for Rational Decision Making, Cornell Human Ecology

Ithaca, NY

Undergraduate Researcher

Nov 2024 – Present

- Design and build modular web automation pipelines with PyAutoGUI, Selenium, and Playwright to conduct decision experiments across ChatGPT's and DeepSeek's web interfaces, eliminating manual data entry.
- Collect and analyze 6,000+ rows of experimental data using Pandas, applying t-tests and ANOVAs to identify cognitive bias patterns across AI models when selecting between risky and safe options in fictional scenarios.
- Develop a structured evaluation framework with rubrics for consistency, choice dominance, exhibited bias, and emotional or moral reasoning to assess how LLMs judge the quality of another agent's decisions.
- Contribute to an upcoming research paper on framing effects in decision theory by benchmarking flagship LLMs' susceptibility to established human cognitive biases.